All-On Allocation Model with Overlapping Intervals

1 Circuit

A circuit has N customers, each of class $j \in \{S, M, L\}$ with current I_j (A) and load $P_j = VI_j$ (kW) at voltage V. A composition $\boldsymbol{n} = (n_S, n_M, n_L)$, $n_S + n_M + n_L = N$, occurs with probability

$$w(\boldsymbol{n}) = \frac{N!}{n_S!\, n_M!\, n_L!}\, p_S^{n_S} p_M^{n_M} p_L^{n_L}, \qquad p_S + p_M + p_L = 1. \tag{1}$$

2 Allocation

There are $K \in \{1, 2, 3\}$ free intervals $[s_k, e_k)$, $k = 1, \ldots, K$, which may overlap. Each medium and large customer is assigned exactly one interval. Let m_k and l_k be the medium and large customers in interval k:

$$\sum_{k=1}^{K} m_k = n_M, \qquad \sum_{k=1}^{K} l_k = n_L, \qquad m_k, l_k \in \mathbb{Z}_{\geq 0}. \tag{2}$$

Inside its interval a customer draws its own class current and load; outside it, it behaves as a small customer. Small customers never change.

3 Segments

Only time inside at least one interval is analysed; time outside every interval is ignored. Cut this time at every interval edge. Each resulting segment $g = [a_g, b_g)$ is covered by a non-empty set of intervals $C(g) = \{k : s_k \leq a_g,\ b_g \leq e_k\}$; overlaps give $|C(g)| \geq 2$. During segment g,

$$M_g = \sum_{k \in C(g)} m_k, \qquad L_g = \sum_{k \in C(g)} l_k, \qquad S_g = N - M_g - L_g. \tag{3}$$

The circuit current and the energy consumed in the segment are

$$A_g = I_S S_g + I_M M_g + I_L L_g \quad \text{(A)}, \tag{4}$$

$$E_g = \big(P_S S_g + P_M M_g + P_L L_g\big)\,(b_g - a_g) \quad \text{(kWh)}. \tag{5}$$

4 Solar soak

Let $X_d(t)$ be the VIC1 utility-solar curtailment (MW) on day d, slot t ($\Delta t = 5$ min). If a share u of the N_{VIC} customers join the tariff, $C = uN_{\text{VIC}}/N$ circuits share it, so one circuit has

$$x_d(t) = X_d(t)/C. \tag{6}$$

Curtailment is what remains after existing demand, so only load added by the free intervals can soak it. In slot $t \in g$ this is

$$D(t) = (P_M - P_S)\, M_g + (P_L - P_S)\, L_g \quad \text{(kW)}. \tag{7}$$

The solar soak per circuit per day, over the days $\mathcal{D}$ (one chosen day, or all 365 days of 2025), is

$$\text{soak} = \frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} \sum_{t} \min\big(D(t),\, x_d(t)\big)\, \Delta t \quad \text{(kWh/day)}. \tag{8}$$

The minimum is taken day by day before averaging; since min is concave, using the mean curtailment profile instead would overstate soak.

Hourly outputs of the single-circuit run sum a kW series f over the slots in each bin B: $\sum_{t \in B} f(t)\, \Delta t$, with f the consumed load $NP_S + D(t)$, the added load $D(t)$, the curtailment share $x_d(t)$, or the soak $\min(D, x_d)$, each averaged over $\mathcal{D}$.

5 Overload

A circuit with composition $\boldsymbol{n}$ and allocation $(\boldsymbol{m}, \boldsymbol{l})$ overloads if the current exceeds the limit T in any segment:

$$r(\boldsymbol{n}, \boldsymbol{m}, \boldsymbol{l}) = \begin{cases} 1, & \max_g A_g > T, \\ 0, & \text{otherwise.} \end{cases} \tag{9}$$

6 Auto allocation and violation probability

The auto allocation minimises the peak current,

$$(\boldsymbol{m}^*, \boldsymbol{l}^*)(\boldsymbol{n}) = \arg\min_{\boldsymbol{m}, \boldsymbol{l}} \max_g A_g, \tag{10}$$

checking every integer split (ties: the first split found). If every circuit uses its auto allocation, the probability that a circuit overloads is

$$\mathbb{P}(\text{violation}) = \sum_{\boldsymbol{n}} w(\boldsymbol{n})\, r\big(\boldsymbol{n}, \boldsymbol{m}^*(\boldsymbol{n}), \boldsymbol{l}^*(\boldsymbol{n})\big), \tag{11}$$

skipping compositions with $w(\boldsymbol{n}) < 10^{-15}$. A composition has a safe allocation if and only if its auto allocation is safe.

7 Pareto front

Both objectives are averages over compositions. Each circuit may use its own allocation $a(\boldsymbol{n})$:

$$O_{\text{safe}} = 1 - \sum_{\boldsymbol{n}} w(\boldsymbol{n})\, r\big(\boldsymbol{n}, a(\boldsymbol{n})\big) \qquad \text{(share of circuits without overload)}, \tag{12}$$

$$O_{\text{soak}} = \sum_{\boldsymbol{n}} w(\boldsymbol{n})\, \text{soak}\big(\boldsymbol{n}, a(\boldsymbol{n})\big) \qquad \text{(expected soak per circuit, kWh/day)}, \tag{13}$$

with soak averaged over all days of 2025. Both are to be maximised.

For each composition every integer allocation is checked, keeping

$$\sigma_{\max}(\boldsymbol{n}) = \max_{a} \operatorname{soak}(\boldsymbol{n}, a), \qquad \sigma_{\text{safe}}(\boldsymbol{n}) = \max_{a:\, r(\boldsymbol{n},a)=0} \operatorname{soak}(\boldsymbol{n}, a). \tag{14}$$

Because $r \in \{0, 1\}$, only these two allocations can lie on the front. If several allocations reach $\sigma_{\max}$, a safe one is preferred. Start with every composition on its $\sigma_{\max}$ allocation (most soak). Composition $\boldsymbol{n}$ can then be moved to its safe allocation, which raises O_{safe} by $w(\boldsymbol{n})$ and lowers O_{soak} by $w(\boldsymbol{n})\,\Delta(\boldsymbol{n})$, where

$$\Delta(\boldsymbol{n}) = \sigma_{\max}(\boldsymbol{n}) - \sigma_{\text{safe}}(\boldsymbol{n}). \tag{15}$$

Moving compositions in increasing order of $\Delta(\boldsymbol{n})$ traces the Pareto front; the segment between two consecutive points is reached by moving only part of the circuits with that composition. The front ends at $O_{\text{safe}} = 1 - \mathbb{P}(\text{violation})$ of the auto allocation.

Along the front the share of medium and large customers placed in interval k is

$$\frac{\sum_{\boldsymbol{n}} w(\boldsymbol{n}) \left(m_k + l_k\right)\left(a(\boldsymbol{n})\right)}{\sum_{\boldsymbol{n}} w(\boldsymbol{n})\,(n_M + n_L)}. \tag{16}$$

With one interval there is a single allocation, so the front is one point. Every allocation with fewer intervals is also available with more when the interval sets are nested, so the front for $K + 1$ intervals is never below the front for K.

The uptake u sets $x_d(t) = X_d(t)/C$: at low uptake each circuit has more curtailment than added load, so soak is limited by load; at high uptake curtailment is the limit and spreading load over more hours soaks more.

8 Computation

Soak depends on an allocation only through (M_g, L_g) in each segment, so for every segment a table is precomputed once:

$$F_g(M, L) = \frac{1}{|\mathcal{D}|} \sum_{d} \sum_{t \in g} \min\big((P_M - P_S)M + (P_L - P_S)L,\ x_d(t)\big)\,\Delta t, \tag{17}$$

$$\operatorname{soak}(\boldsymbol{n}, a) = \sum_{g} F_g(M_g, L_g). \tag{18}$$

Each composition has $\binom{n_M+K-1}{K-1}\binom{n_L+K-1}{K-1}$ allocations; all are evaluated (9.4×10^6 over all compositions for $K = 3$).

Quantity	Code (`play/`)
Inputs, $P_j = VI_j$, C	`config.py`: `load_kw`, `circuits`
$x_d(t)$	`solar.py`: `excess_per_circuit`
Segments, S_g, M_g, L_g, A_g, $w(\boldsymbol{n})$	`circuit.py`: `segments`, `active`, `peak`, `weight`
Auto allocation, $\mathbb{P}$(violation)	`circuit.py`: `auto_allocation`, `violation_probability`
$D(t)$, hourly energy, F_g	`soak.py`: `load_profiles`, `kwh`, `soak_table`
$\sigma_{\max}, \sigma_{\text{safe}}$, front	`frontier.py`: `evaluate`, `pareto_front`
Single-circuit run, Pareto run	`run_allocation.py`, `run_pareto.py`

9 Inputs

Shared inputs are in `play/config.py`; K, $\boldsymbol{n}$, the allocation and the day are set per run in `play/run_allocation.py`, and the uptakes compared in `play/run_pareto.py`.

Symbol	Code variable	Default
N	`N`	40
T	`T`	1000 A
I_S, I_M, I_L	`CURRENT`	20, 40, 100 A
V	`V`	240 V
u	`UPTAKE`	0.1
N_{VIC}	`VIC_CUSTOMERS`	2,900,000
$\mathcal{D}$	`SOLAR_DAY`	2025-11-06 (Pareto: all of 2025)
p_S, p_M, p_L	`P_TYPE`	0.69, 0.30, 0.01
$[s_k, e_k)$	`INTERVAL_SETS`	11–14 / 8–11, 11–14 / 8–11, 10:30–13:30, 11–14
K	`NUM_INTERVALS`	3
$\boldsymbol{n}$	`COMPOSITION`	(32, 7, 1)
$(\boldsymbol{m}, \boldsymbol{l})$	`ALLOCATION`	auto